

For more sophisticated filters. Classic analogue filters such as Butterworth, Chebyshev or elliptic can be discretized using s -to- z transformation methods. The “translation” is approximate, e.g., the Nyquist frequency limit.

- ① **bilinear** transform
- ② **impulse-invariant**
- ③ **pole-zero matching** technique

We have to make the transition from a continuous signal transfer function to the discretized signal transfer function

$$H_a(s) \Rightarrow H_d(z)$$

i.e. we need to do an $s(z)$ s -plane-to- z -plane mapping:

$$H_d(z) = H_a[s(z)] .$$

Design of IIR filters: the bilinear transform

For mapping the s plane to the z plane we use the transformation

$$s = \frac{2}{T} \frac{z - 1}{z + 1}$$

Explanation

We look for a simple approximation of $\ln x$.

The Taylor series expansion around zero (MacLaurin series) is not feasible as neither $\ln x$ nor its derivatives can be evaluated at this point. Instead we do the series expansion around one or equivalently the expansion of $\ln(1 \pm x)$ around zero.

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^n \frac{x^n}{n} + \dots,$$

$$\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots - \frac{x^n}{n} - \dots$$

where the series are convergent on the $-1 < x \leq 1$ interval and the convergence is faster for smaller $|x|$.

$$\ln(1-x) - \ln(1+x) = \ln \frac{1-x}{1+x} = -2 \left(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \right).$$

Applying the change of variables

$$\frac{1-x}{1+x} = z, \quad x = \frac{1-z}{1+z} \quad \rightarrow \quad \ln z = -2 \left[\frac{1-z}{1+z} + \frac{1}{3} \left(\frac{1-z}{1+z} \right)^3 + \frac{1}{5} \left(\frac{1-z}{1+z} \right)^5 + \dots \right], \quad 0 < z < +\infty,$$

The advantages of the above series expansion:

- i.) covers the entire $(0, +\infty)$ domain of the logarithm function;
- ii.) easy to handle it numerically as it is a simple power series of the variable $x = \frac{1-z}{1+z}$;
- iii.) fast convergent due to the absence of the even powered terms.

Therefore the first order approximation of $\ln z$ is

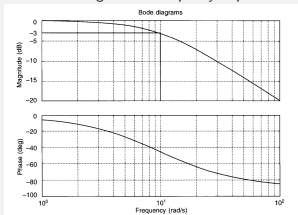
$$\ln z \approx 2 \frac{z-1}{z+1}, \quad \text{whence}$$

$$\text{if } z = e^{sT} \rightarrow s = \frac{1}{T} \ln z \rightarrow s \approx \frac{2}{T} \frac{z-1}{z+1}.$$

Example: first-order lowpass filter with a cut-off angular frequency $\omega_c = 10$ rad/s, ($f_s = 16$ Hz)

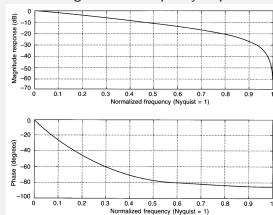
$$H_a(s) = \frac{1}{1 + s/\omega_c} = \frac{\omega_c}{s + \omega_c} \Rightarrow H_d(z) = H_a\left(2f_s \frac{z-1}{z+1}\right) = \frac{\omega_c}{\omega_c + 2f_s} \frac{z+1}{z + \frac{\omega_c - 2f_s}{\omega_c + 2f_s}} = \frac{0.238(z+1)}{z - 0.524}.$$

Analogue filters frequency response



Cut off frequency of 1.6 Hz

Digital filters frequency response



Cut off frequency of 1.5 Hz

If we are not happy with the similarity between the discretized and analogue version we do a **pre-warping** of the frequencies

Pre-warping

Modifying the s -domain transfer function in a particular way such that the digital filter's gain at a 'frequency of interest' (e.g. cut-off) is similar to that of the analogue filter while the direct current gain remains unchanged:

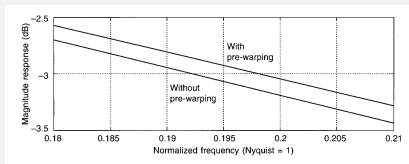
$$\omega'_c = \frac{2}{T} \tan \left(\frac{\omega_c T}{2} \right) .$$

Explanation

$$H_d(e^{i\omega T}) = H_a\left(\frac{2}{T} \frac{e^{i\omega T} - 1}{e^{i\omega T} + 1}\right) = H_a\left(i \frac{2}{T} \tan(\omega T/2)\right) = H_a(i\omega')$$

Example cont.: first-order lowpass filter with a cut-off angular frequency $\omega_c = 10$ rad/s, ($f_s = 16$ Hz)

$$\omega'_c = 2f_s \tan\left(\frac{\omega_c}{2f_s}\right) = 32 \tan\left(\frac{10}{32}\right) \approx 10.34 \text{ rad/s} \rightarrow H'_d(z) = \frac{\omega'_c}{\omega'_c + 2f_s} \frac{z + 1}{z + \frac{\omega'_c - 2f_s}{\omega'_c + 2f_s}} = \frac{0.244(z + 1)}{z - 0.512}.$$



Filter design steps

- 1 Prewarp the cutoff frequencies
- 2 Design the necessary analog filter
- 3 Apply the bilinear transform to the transfer function
- 4 Normalize the resultant transfer function to be monotonic and have a unity passband gain (0 dB).

Note!

- 1 Before doing the bilinear transformation check whether pre-warping produces significant change in the frequency breakpoint. If ω'_c differs from ω_c by less than 1% then pre-warping is probably not worth it.
- 2 Sufficiently high sampling rate makes pre-warping unnecessary.
- 3 Pre-warping ensure matching gains (digital vs analogue) for a single breakpoint (and 0 Hz). For higher order filters with several breakpoints we have to pick the one we prefer.

Exercise

- 1 Derive up to second order the pre-warp correction (use the MacLaurin series).
- 2 In the previous example estimate the correction in ω_c for a sampling rate of 100 Hz.

Design of IIR filters: the impulse-invariant method

Recap: response function of LTI systems

ANALOGUE

$$y(t) = h(t) \star x(t) ,$$

$x(t)$ - input, $h(t)$ - impulse response, $y(t)$ - output.
Applying Laplace-transform:

$$Y(s) = H(s) \cdot X(s) .$$

If $x(t) = \delta(t)$

$$y(t) = h(t) \star \delta(t) = h(t) .$$

Applying Laplace-transform:

$$Y(s) = H(s) \cdot 1 = H(s) .$$

DIGITAL

$$y[n] = h[n] \star x[n]$$

$x[n]$ - input, $h[n]$ - impulse response, $y[n]$ - output.
Applying z-transform:

$$Y(z) = H(z) \cdot X(z) .$$

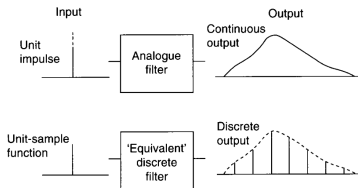
If $x[n] = \delta[n]$

$$y[n] = h[n] \star \delta[n] = h[n]$$

Applying z-transform:

$$Y(z) = H(z) \cdot 1 = H(z) .$$

$$H(s) \xrightarrow{\mathcal{L}^{-1}} h(t) \xrightarrow{\text{sampling}} h[n] \xrightarrow{z\text{-transform}} H(z) \rightarrow \text{frequency response}$$



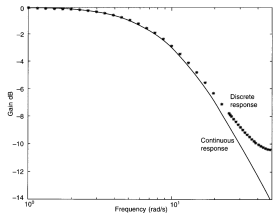
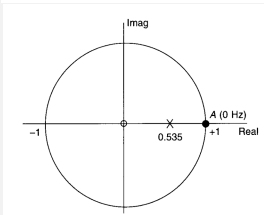
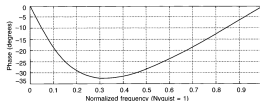
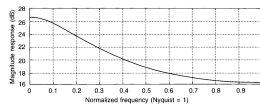
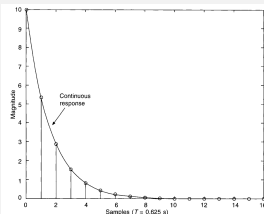
Example: first-order lowpass filter with a cut-off angular frequency $\omega_c = 10 \text{ rad/s}$, ($f_s = 16 \text{ Hz}$)

$$H_a(s) = \frac{\omega_c}{s + \omega_c} \xrightarrow{\mathcal{L}^{-1}} h(t) = \omega_c \cdot e^{-\omega_c t} \xrightarrow{\text{sampling}} h[n] = h(nT) = \omega_c \cdot e^{-\omega_c nT} = \omega_c a^n, \quad a \equiv e^{-\omega_c T}$$

$$\xrightarrow{\text{z-transform}} H_d(z) = \frac{\omega_c z}{z - a} = \frac{10z}{z - 0.535}$$

In order to keep the d.c. gain ($\omega = 0$, i.e., $z = 1$) at unity we have to normalize the above by dividing it by $H_d(1)$:

$$H_d(1) = \frac{10}{1 - 0.535} = \frac{10}{0.465} \rightarrow H_d(z) = \frac{0.465z}{z - 0.535}$$



Exercise

Prove that the above figures are correct. Derive the formulae and represent them graphically.

Convert the poles and zeros in the s -plane into corresponding poles and zeros in the z -plane using the $z = e^{sT}$ relationship.

Example: low-pass analogue filter with pole in σ

$$H_a(s) = \frac{\sigma}{s + \sigma} \rightarrow G_a(\omega) = \frac{\sigma}{\sqrt{\omega^2 + \sigma^2}}$$

Single pole at σ in the s -plane $\rightarrow$ a single pole in $a = e^{\sigma T}$ in the z -plane. $\rightarrow H_d(z) = \frac{k}{z - a}$,

We expect the gain at $\omega = 0$ to be unity:

$$G_d(\omega) = \frac{k}{|e^{j\omega T} - a|} \rightarrow G_d(0) = 1 = \frac{k}{|1 - a|} \rightarrow k = |1 - a|.$$

$$\rightarrow H_d(z) = \frac{|1 - a|}{z - a}, \quad a = e^{\sigma T}.$$

Problem: the only frequency where we control the gain is at $\omega = 0$. While $G_a(\infty) = 0$ the the digital filter's gain at the Nyquist frequency ($z = -1$) will vanish only in the $T \rightarrow 0$ very high resolution limit.

Example continued: low-pass analogue filter with pole in σ

$$G_d(\omega_N) = \left| \frac{1 - a}{1 + a} \right| \in (0, 1) \quad \text{for } T \in (0, \infty).$$

We fix it by adding a zero at -1 ($G_d(\omega_N) = 0$).

Example continued: low-pass analogue filter with pole in σ

$$H_d(z) = k \frac{z+1}{z-a} \rightarrow G_d(0) = 1 = k \frac{2}{|1-a|} \rightarrow k = \frac{|1-a|}{2} \rightarrow H_d(z) = \frac{|1-a|}{2} \frac{z+1}{z-a}.$$

Sometimes we need to equalize the gains of the two filters at some frequency between 0 and ω_N .

Example: high-pass analogue filter with pole in σ . Equalize the gain of the analogue and digital filters at $\sigma \ll \omega_0 < \omega_N$.

$$H_a(s) = \frac{s}{s+\sigma} \rightarrow G_a(0) = 0, \quad G_a(\omega_0) = \frac{\omega_0}{\sqrt{\omega_0^2 + \sigma^2}} < 1$$

$$H_d(z) = k \frac{z-1}{z-a} \rightarrow G_d(0) = 0, \quad G_d(\omega_0) = k \frac{e^{j\omega_0 T} + 1}{e^{j\omega_0 T} - e^{j\sigma T}} = \frac{\omega_0}{\sqrt{\omega_0^2 + \sigma^2}} \rightarrow k = \frac{\omega_0}{\sqrt{\omega_0^2 + \sigma^2}} \frac{e^{j\omega_0 T} - e^{j\sigma T}}{e^{j\omega_0 T} + 1}$$

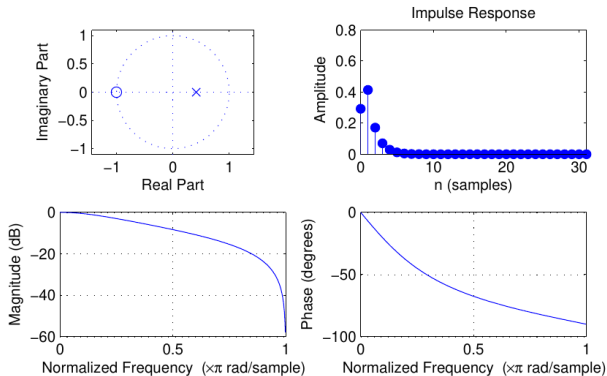
Try for $\sigma = 1$ rad/s, $\omega_0 = 10$ rad/s, $f_s = 5$ Hz and show that $H_d(z) = 0.91 \frac{z-1}{z-82}$.

For high-pass filters it is easier to do the equalization at the Nyquist frequency. However, for bandpass filters it has to be done at an intermediate frequency.

- ① **Butterworth filters** Have no ripples, gain falls monotonically across the pass and transition band. Within the passband, the gain drops slowly to -3 dB. Outside it drops by 2^N per octave.
- ② **Chebyshev type I filters** Distribute the gain error uniformly throughout the passband (equiripples) and drop off monotonically outside.
- ③ **Chebyshev type II filters** Distribute the gain error uniformly throughout the stopband (equiripples) and drop off monotonically in the passband.
- ④ **Elliptic filters** Distribute the gain error as equiripples both in the passband and stop- band. Optimal in terms of the combination of the passband-gain tolerance, stopband-gain tolerance, and transition-band width .

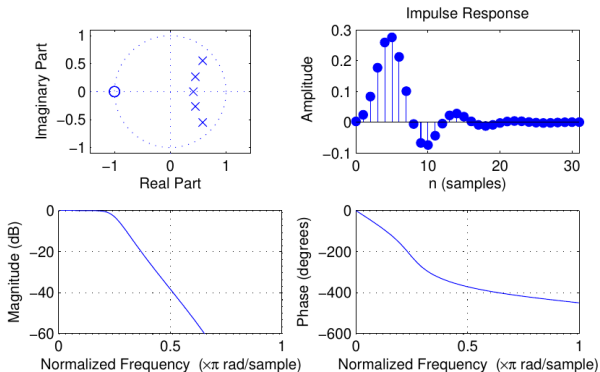
Butterworth filter design example

order: 1, cut-off frequency: $0.25 f_N$



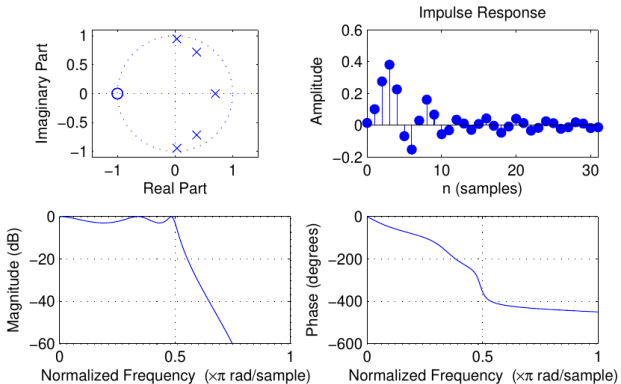
Butterworth filter design example

order: 5, cut-off frequency: $0.25 f_N$



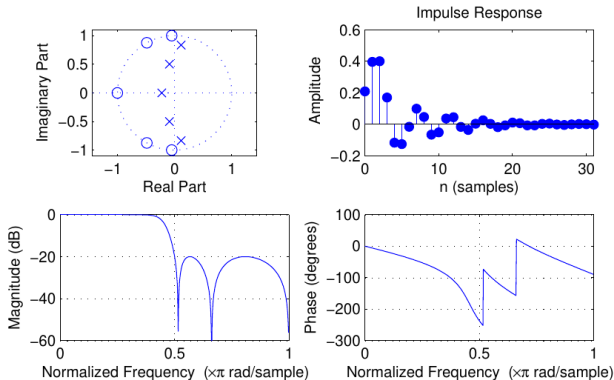
Chebyshev type I filter design example

order: 5, cut-off frequency: $0.5 f_N$, pass-band ripple: -3 dB



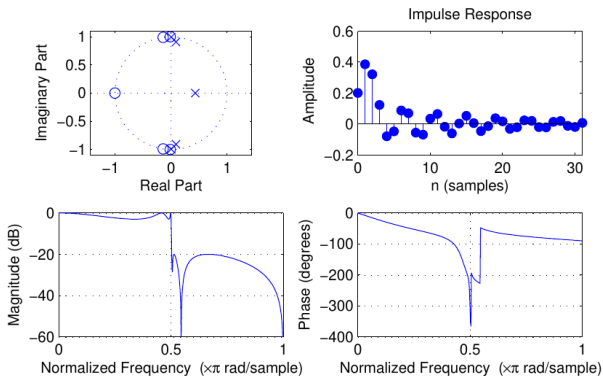
Chebyshev type II filter design example

order: 5, cut-off frequency: $0.5 f_N$, pass-band ripple: -20 dB



Elliptic filter design example

order: 5, cut-off frequency: $0.25 f_N$, pass-band ripple: -3 dB, stop-band ripple: -20 dB



Advantages of FIR filters

- ➊ **Always stable:** FIR filters have no feedback poles, so their impulse response is finite.
- ➋ **Exact linear phase is possible:** if the impulse response is symmetric or antisymmetric, the filter has linear phase.
- ➌ **No phase distortion:** linear phase means all frequency components are delayed by the same amount.
- ➍ **Simple nonrecursive implementation:**

$$y[n] = \sum_{k=0}^M b_k x[n - k].$$

- ➎ **Easy to design for arbitrary magnitude responses:** windowing, frequency-sampling, and optimization methods can be used.
- ➏ **Insensitive to coefficient quantization:** FIR filters are usually numerically more robust than IIR filters.

Disadvantages of FIR filters

- ➊ **Higher order is often needed:** for a sharp transition band, FIR filters usually need more coefficients than IIR filters.
- ➋ **More computations:** a high-order FIR filter requires many multiplications and additions per sample.
- ➌ **More memory:** many past input samples and coefficients must be stored.
- ➍ **Larger delay:** linear-phase FIR filters introduce a group delay of approximately

$$\frac{M}{2}$$

samples.

- ➎ **Less natural for analog-prototype design:** classical analog filters such as Butterworth or Chebyshev lead more directly to IIR filters.

Two classes of methods:

- 1 'Fourier' or 'windowing' method;
- 2 frequency sampling method.

Theorem

For a linear phase response the coefficients of the FIR filter must be symmetric.

Example: running average filters

$$y[n] = \frac{1}{4} (x[n] + x[n-1] + x[n-2] + x[n-3])$$

or

$$y[n] = \frac{1}{5} (2x[n] + x[n-1] + 3x[n-2] + x[n-3] + 2x[n-4])$$

$$Y(z) = X(z) \frac{1}{5} (2 + z^{-1} + 3z^{-2} + z^{-3} + 2z^{-4}) = X(z)H(z),$$

$$H(z) = \frac{2z^4 + z^3 + 3z^2 + z + 2}{5z^4}$$

Proof

$$H(z) = \sum_{n=0}^{N-1} h[n]z^{-n} \quad \text{where} \quad h[n] = h[N-1-n].$$

$$\rightarrow H(e^{i\omega}) = \frac{1}{2} \left\{ \sum_{n=0}^{N-1} h[n]e^{-ni\omega} + \sum_{n=0}^{N-1} h[N-1-n]e^{-ni\omega} \right\}, \quad n \rightarrow N-1-n \text{ in the second sum}$$

$$\rightarrow H(e^{i\omega}) = \frac{1}{2} \left\{ \sum_{n=0}^{N-1} h[n]e^{-ni\omega} + \sum_{n=0}^{N-1} h[n]e^{-(N-1-n)i\omega} \right\} =$$

$$= e^{i\omega \frac{N-1}{2}} \frac{1}{2} \left\{ \sum_{n=0}^{N-1} h[n]e^{-(n-\frac{N-1}{2})i\omega} + \sum_{n=0}^{N-1} h[n]e^{-(\frac{N-1}{2}-n)i\omega} \right\} = e^{i\omega \frac{N-1}{2}} \sum_{n=0}^{N-1} h[n] \cos \left[\left(n - \frac{N-1}{2} \right) \omega \right]$$

$$\angle H(e^{i\omega}) = \angle e^{i\omega \frac{N-1}{2}} + \angle \sum_{n=0}^{N-1} h[n] \cos \left[\left(n - \frac{N-1}{2} \right) \omega \right] = \angle e^{i\omega \frac{N-1}{2}} = \omega \frac{N-1}{2} \sim \omega \quad \square$$

Fourier/window method for low-pass FIR filters

Starting from a continuous signals spanning the whole real axis:

$$y(t) = \mathcal{H}[x(t)] = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau = (x * h)(t)$$

$$Y(f) = X(f) \cdot H(f)$$

For an ideal **low-pass filter with cut-off frequency** f_c , $H(f) = \text{rect}(f/2f_c) \rightarrow$ the impulse response is $h(t) = \mathcal{F}[H(f)] = 2f_c \text{sinc}_{\pi}(2tf_c)$

Reminder

$$\mathcal{F}^{-1}[\text{rect}(f/f_s)] = \frac{1}{T} \text{sinc}_{\pi}(t/T) \quad \text{where } T = 1/f_s$$

$$y(t) = (x * h)(t) \quad \rightarrow \quad \text{after sampling } \hat{y}(t) = (\hat{x} * \hat{h})(t) .$$

$$\begin{aligned} \hat{x}(t) &= x(t) \cdot \Pi_T(t) \\ \mathcal{F}[\hat{x}] &= T \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} x(t) \delta(t - nT) e^{i\omega t} dt = T \sum_{n=-\infty}^{\infty} x[n] e^{-i\omega nT} = T \cdot X(z) , \quad z \equiv e^{-i\omega T} \end{aligned}$$

Similarly

$$\mathcal{F}[\hat{y}] = T \cdot Y(z) , \quad \mathcal{F}[\hat{h}] = T \cdot H(z)$$

$$Y(z) = T \cdot X(z) \cdot H(z) \quad \rightarrow \quad y[n] = T \cdot h[n] * x[n]$$

So the coefficients in the discrete convolution should be $T \times$ the samples $h(nT)$.

If using digital signal sampled at T intervals the filter coefficients to use in

$$y[k] = \sum_{n=0}^a$$

$$h_n = T \cdot h[n] = \frac{1}{n\pi} \sin \left(n\pi \frac{f_c}{f_N} \right)$$

What values should n take?

① n takes an infinity of $\rightarrow$ computationally not feasible.

② n takes negative values $\rightarrow$ non-causal filter.

We fix it by

① Apply a cutoff (finite, symmetric temporal window) $\rightarrow -N/2 \leq n \leq N/2$;

② Shift the window by half of its size so that $n \geq 0 \rightarrow a_n \equiv h_{n-N/2}$.

$$y[k] = \sum_{n=0}^N a_n x[k-n] .$$

$$H(z) = \frac{\sum_{n=0}^N a_n z^n}{z^N}$$

Note! Typically the number of sample points is odd, i.e., there is a central point and two symmetrically positioned “wings”. If the order of the filter is N (even), the number of samples is $N + 1$. If N is odd then the number of samples is N .

Exercise

Design a FIR filter of order 20 with cutoff frequency of 4kHz and sampling frequency of 20kHz.

The sinc_π impulse response corresponding to the rectangle function in frequency space is a relatively slowly decaying function

→ for a finite order N we “truncate” it significantly (equivalent to spectral leakage)

→ the frequency response is not optimal (low width of the main lobe, large peak of the side lobe typical for the rectangle window)(see the “Effect of windowing” in dsp05.pdf)

For a better result we use a (smoothing) **taper window** w_n multiplicatively over a_n which improves the frequency response:

$$y[n] = \tilde{a}[n] * x[n] , \quad \tilde{a}[n] = a[n] \cdot w[n]$$

This is equivalent to applying the window on each input signal segment of length N that is employed for getting a new output value.

Fourier/window method for high-pass, band-pass and band-stop FIR filters

For the low-pass filter we kept the frequencies within $(-f_c, f_c)$ and suppressed the rest applying the rectangle function in the frequency space.

Similarly we can keep-

/suppress other regions of the frequency space by combining rectangle functions:

Filter-type	Window (frequency)	Impulse response (cont. time)	Impulse response (disc. time)
low-pass	$\text{rect}(f/2f_c)$ 	$h(t; f_c) \equiv 2f_c \text{sinc}_\pi(2tf_c)$	$a[n; f_c] \equiv \frac{1}{n\pi} \sin\left(n\pi \frac{f_c}{f_N}\right)$
high-pass	$1 - \text{rect}(f/2f_c)$ 	$\delta(t) - h(t; f_c)$	$\delta[n] - a[n; f_c]$
band-pass	$\text{rect}(f/2f_h) - \text{rect}(f/2f_l)$ 	$h(t; f_h) - h(t; f_l)$	$a[n; f_h] - a[n; f_l]$
band-stop	$1 - \text{rect}(f/2f_h) + \text{rect}(f/2f_l)$ 	$\delta(t) - h(t; f_h) + h(t; f_l)$	$\delta[n] - a[n; f_h] + a[n; f_l]$